Exercise 1

a) Proof by induction: $P(n) = \text{"}2^n < n!\text{"}$

Base step: $2^4 = 16 < 24 = 4!$

Inductive step: Suppose $2^n < n!$, since $n \geq 4$ we also have $2 < n+1$. Multiplying the two inequalities we get $2^{n+1} < (n+1)!$.

b) Fix $\varepsilon > 0$.

By the Archimedean principle $\exists\, N \in \mathbb{N}$ s.t. $\frac{1}{N} < \varepsilon$

From a) we know that $\frac{2^n}{n!} < 1$, Hence for $n \geq N$, $n \geq 4$ we have

$\frac{2^n}{(n+1)!} = \frac{2^n}{n!} \cdot \frac{1}{n+1} < 1 \cdot \varepsilon < \varepsilon$, as required in the definition of limit

Exercise 2

a) i) $\max(A) = \sup(A) = 1$

$\inf(A) = 0$, A has no min

ii) $\sup(A) = +\infty$, A has no max

$\inf(A) = \min(A) = 0$

b) $\limsup_{n \to +\infty} a_n = 3$, $\liminf_{n \to +\infty} a_n = 1$

acc points $= \{1, 3\}$

$\left(a_n = \frac{1}{n+1} + 2 + (-1)^n\right)$

Exercise 3

a) i) The limit does not exist, indeed:

Take $n_k = 2k$

$$\lim_{k\to+\infty} \frac{n_k^2 \cos(n_k)}{n_k+2} = \lim_{k\to+\infty} \frac{4k^2}{2k+2} = \lim_{k\to+\infty} k\left(\frac{4}{2+2/k}\right) = +\infty \cdot \frac{4}{2+0} = +\infty$$

and $\tilde{n}_k = 2k+1$

$$\lim_{k\to+\infty} \frac{\tilde{n}_k^2 \cos(\tilde{n}_k)}{\tilde{n}_k+2} = \lim_{k\to+\infty} \frac{-4k^2 - 4k - 1}{2k+3} = \lim_{k\to\infty} k\left(\frac{-4-4/k-1/k^2}{2+3/k}\right) = -\infty$$

Since the two subsequences have different limits the claim is proved.

ii) Let us first compute

$$\lim_{n\to\infty} \cancel{\sin}\left(\frac{2n+\pi}{n^2}\right) = \lim_{n\to\infty} \frac{2}{n} + \frac{\pi}{n^2} = \frac{2}{+\infty} + \frac{\pi}{+\infty} = 0.$$

Since sin is continuous at 0 we can write

$$\lim_{n\to+\infty} \sin\left(\frac{2n+\pi}{n^2}\right) = \sin\left(\lim_{n\to+\infty} \frac{2n+\pi}{n^2}\right) = \sin(0) = 0$$

b) i) We only care for $x<5$, hence $|x-5| = 5-x$

$$\lim_{x\to5^-} \frac{x^2-25}{|x-5|} = \lim_{x\to5^-} \frac{\cancel{(x-5)}(x+5)}{\cancel{(5-x)}} = \lim_{x\to5^-} -(x+5) = -10$$

ii) Using Taylor expansion with Peano reminder, centered at 0,

should be -1/2

$$\lim_{x\to0} \frac{\cos(x)-1}{x(e^x-1)} = \lim_{x\to0} \frac{\frac{1}{2}x^2 + o(x^2)}{x(x+o(x))} = \frac{1}{2}\,\frac{1+o(1)}{1+o(1)} = \frac{1}{2}$$

c) $$\lim_{x\to+\infty} \frac{\int_0^x e^{t^2}dx}{e^{x^2}} =$$

$$= \lim_{x\to+\infty} \frac{e^{x^2}}{2x e^{x^2}}$$

$$= \lim_{x\to+\infty} \frac{1}{2x} = 0$$

we use de l'Hopital, noticing that $\lim_{x\to+\infty} \int_0^x e^{t^2}dx \geq \lim_{x\to+\infty} \int_0^x 1\,dx \geq \lim_{x\to+\infty} x = +\infty$ and $\lim_{x\to+\infty} e^{x^2} = +\infty$ and that both functions are differentiable. $\left(\int_0^x e^{t^2}dx\right)' = e^{x^2}$ by fundamental thm of calculus

Exercise 4

a) i) We know from class that
$\sum_{k=0}^{\infty} \frac{x^k}{k!}$ is convergent $\forall x$ and the result is e^x.
Hence $\sum_{k=0}^{\infty} \frac{2^k}{k!} = e^2$.

ii) This is a geometric series. It is convergent since $\frac{1}{1+\pi^2} < 1$. Notice that the first term is missing, other than this we can apply the usual formula

$$\sum_{k=1}^{\infty} (1+\pi^2)^{-k} = \sum_{k=0}^{\infty} (1+\pi^2)^{-k} - 1 = \frac{1}{1-\frac{1}{1+\pi^2}} - 1 = \frac{1}{\pi^2}$$

b) We look at 3 different cases:

$a<0$

$\sum_{n=1}^{\infty} \frac{(-1)^n}{n} \left(\frac{1}{e^a}\right)^n$ is divergent, and absolutely divergent.
In fact the terms of the sequence don't converge to 0. One way to check this is to notice that their absolute values, $\left(\frac{1}{e^a}\right)^n \cdot \frac{1}{n}$, go to $+\infty$. This is true since $\frac{1}{e^a} > 1$, and the exponential grows faster than n.

$a=0$

$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ is convergent, but absolutely divergent.
It converges thanks to Leibnitz criterion: $\frac{1}{n}$ is positive, decreasing and it converges to 0. Moreover the sign is alternating thanks to $(-1)^n$.
The series of absolute values is $\sum_{n=1}^{\infty} \frac{1}{n}$, which we know to be divergent from class.

$a>0$

$\sum_{n=1}^{\infty} \frac{(-1)^n}{n} \left(\frac{1}{e^a}\right)^n$ is absolutely convergent and hence convergent
Indeed, the series of absolute value is $\sum_{n=1}^{\infty} \frac{1}{n} \left(\frac{1}{e^a}\right)^n$.
Since it has positive terms we can apply the ratio test and compute

$$\lim_{n\to+\infty} \frac{n}{n+1} \frac{(1/e^a)^{n+1}}{(1/e^a)^n} = \lim_{n\to+\infty} \frac{n}{n+1} \frac{1}{e^a} = \frac{1}{e^a} < 1$$

and deduce that the series of absolute values is convergent

Exercise 5

	1	2	3	4	5	6	7	8	9	10
left	T	T	F	T	F	F	F	F	F	F
right	F	F	T	F	T	T	F	T	T	F

Exercise 6

a) For $x \neq 0$ $\frac{1}{x}$ is differentiable, and x^2 and $\cos(x)$ are differentiable for all $x \in \mathbb{R}$.
Hence for $x \neq 0$ $f(x)$ is differentiable being composition and product of differentiable function
For $x = 0$ we check the definition of differentiable, by computing

$$\lim_{x\to 0} \frac{f(x)-f(0)}{x} = \lim_{x\to 0} \frac{x^3 \sin(1/x)}{x} = \lim_{x\to 0} x^2 \sin(1/x)$$

Now we notice that $-x^2 \leq \sin(1/x)\cdot x^2 \leq x^2$ and conclude that

$$\lim_{x\to 0} x^2 \sin(1/x) = \lim_{x\to 0} -x^2 = \lim_{x\to 0} x^2 = 0 \text{ by sandwich theorem}$$

Hence $f'(0) = 0$.
We compute the derivative of $f(x)$ for $x \neq 0$ with the usual rules of calculus, getting:

$$f'(x) = \begin{cases} 3x^2 \sin(1/x) + x\cos(1/x) & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$$

b) Assume $x > 0$, if $x < 0$ the proof is analogous.
Since g is C^2 on $[0, x]$ we can apply Taylor expansion with Lagrange reminder centered at 0 at 2nd order

$$g(x) = g(0) + \cancel{g'(0)}\cdot(x-0) + \frac{1}{2} g''(x_0)(x-0)^2 \quad \text{since } g'(0) = 0$$

$$= \frac{1}{2} g''(x_0) x^2 + g(0)$$

$$2|g(x) - g(0)| = |g''(x_0)| \cdot |x^2| \leq |x|^2.$$

c) $h : \mathbb{R} \to \mathbb{R}$ is a C^2 function.
Since $h(x_0) = h(x_1)$ by Rolle $\exists\, y_0 \in (x_0, x_1)$ s.t. $h'(y_0) = 0$
Since $h(x_1) = h(x_2)$ by Rolle $\exists\, y_1 \in (x_1, x_2)$ s.t. $h'(y_1) = 0$
$h' : \mathbb{R} \to \mathbb{R}$ is C^1, then since $h'(y_0) = h'(y_1)$
then by Rolle $\exists\, y \in (y_0, y_1)$ s.t. $h''(y) = (h'(y))' = 0$.

Exercise 7

a) i) $\int_1^5 \frac{e^{-\sqrt{x-1}}}{\sqrt{x-1}}\,dx = \int_0^4 \frac{e^{-\sqrt{t}}}{\sqrt{t}}\,dt$ $\qquad x-1=t \quad dx=dt$, $\sqrt{t}=\lambda \quad d\lambda = \frac{1}{2\sqrt{t}}dt$

$= \int_0^2 2e^{-\lambda}\,d\lambda = 2\left[-e^{-\lambda}\right]_0^2 = 2(1-e^{-2})$

ii) $\int_0^{\pi} x\cos\left(\frac{x}{2}\right)dx = \left[2x\sin\left(\frac{x}{2}\right)\right]_0^{\pi} - 2\int_0^{\pi}\operatorname{sen}\left(\frac{x}{2}\right)dx$

$= 2\pi\sin\left(\frac{\pi}{2}\right) - 2\left[2\cos\frac{\pi}{2} - 2\cos 0\right]$

$= 2\pi - 4$

iii) $\int \frac{x^2+2x}{x^2+1}\,dx = \int 1\,dx - \int \frac{1}{x^2+1}dx + \int \frac{2x}{x^2+1}\,dx$ $\qquad x^2+1=t$, $2x\,dx = dt$

$= x - \arctan(x) + \int \frac{1}{t}\,dt + c$

$= x - \arctan(x) + \log(x^2+1) + c$

b) Suppose by contraddiction that $\lim_{x\to+\infty} f(x)$ is not 0.
This means that there exists $\varepsilon>0$ and $x_n \xrightarrow{n} +\infty$ s.t. $f(x_n) > \varepsilon$. We may suppose that x_n is increasing and that $x_{n+1} - x_n > 1$.
We apply the definition of uniform continuity for $\varepsilon/2$, and we get the existence of $\delta>0$ s.t. $|f(x)-f(y)| \le \varepsilon/2$ whenever $|x-y|<\delta$. (We may reduce δ to have $\delta < 1/2$).
This implies, by triangular inequality that $f(x) > \varepsilon/2$ whenever $x \in (x_n-\delta, x_n+\delta)$ $\forall\, n$.
Notice that the intervals $(x_n-\delta, x_n+\delta)$ are all disjoint.
We now compute

$$\int_0^{+\infty} f(x) = \lim_{M\to+\infty}\int_0^M f(x) = \lim_{n\to+\infty}\int_0^{x_n+\delta} f(x)$$

$$\ge \lim_{n\to+\infty}\sum_{k=1}^{n}\int_{x_k-\delta}^{x_k+\delta} f(x) \ge \lim_{n\to+\infty}\sum_{k=1}^{n}\int_{x_k-\delta}^{x_k+\delta}\varepsilon/2$$

$$\ge \lim_{n\to+\infty}(2\delta\cdot\varepsilon/2)\cdot n = +\infty$$

Contraddiction.
Notice that the previous estimates hold true becaus $f(x)\ge 0\ \forall x$.